Comparing Chunking Strategies for Retrieval-Augmented Question Answering

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Abstract

We study how document chunking affects retrieval-augmented question answering in a controlled pipeline where the embedding model, retriever, and generator are held fixed. Four chunking families are compared: fixed-size token windows, recursive chunking, sentence-based chunking, and semantic chunking. Using `Mistral-7B-Instruct-v0.3` served through vLLM on SCC, we find that `recursive_256` is the strongest system on both a 60-question SQuAD v2 subset (60.6 F1) and a 30-question HotpotQA subset (42.6 F1). A parametric-only baseline is substantially weaker, confirming retrieval sensitivity. A fine-grained error analysis of all EM = 0 predictions further shows that 65.7% of SQuAD failures and 72.7% of HotpotQA failures are fixable format or refusal errors rather than genuine retrieval or reasoning errors, pointing to prompt-level improvements as the primary path to higher scores.[1]

1 Introduction

Retrieval-augmented generation (RAG) answers questions by retrieving external evidence and passing it to a language model (Lewis et al., 2020). In practice, retrieval operates over chunks rather than full documents, making chunking a central design choice: it determines what evidence stays together, how likely the retriever is to recover it, and whether the generator receives coherent support or fragmented context.

This project asks: *which chunking strategy maximizes end-to-end QA accuracy in a fixed RAG pipeline, and does the answer differ between local factoid questions and multi-hop questions?*

Formally, let a corpus be D, a question be q, and a chunking function be $c : D \to \{d_1, \dots, d_n\}$. After chunking, each chunk is embedded and indexed. At inference time, the system retrieves the top-K chunks for q, concatenates them into a prompt, and produces an answer $\hat{a}$. The optimization target is end-to-end QA quality:

$$c^* = \arg\max_c \mathbb{E}_{(q,a)} \left[F_1(\hat{a}, a)\right].$$

Retrieval quality and answer quality do not always move together: a system may retrieve locally relevant text yet still fail if the evidence is incomplete or badly fragmented. We therefore evaluate chunking with both end-to-end QA metrics and a post-hoc error analysis that attributes each failure to retrieval, generation format, or genuine model error.

2 Related Work

RAG conditions a language model on retrieved passages rather than relying on parametric memory alone (Lewis et al., 2020). Dense Passage Retrieval (Karpukhin et al., 2020) established bi-encoder dense retrieval as a strong backbone for open-domain QA and serves as our fixed retriever. We evaluate on SQuAD 2.0 (Rajpurkar et al., 2018), which tests single-hop answer extraction, and HotpotQA (Yang et al., 2018), which requires composing evidence across multiple documents, making it a natural pair for studying whether chunking effects depend on question complexity.

Chunking has recently attracted attention as a first-class experimental variable. Smith and Troynikov (2024) find that chunk size and overlap substantially affect retrieval precision and recall, with no single strategy dominating across domains. NVIDIA's end-to-end study (NVIDIA, 2024) shows that retrieval metrics alone can be misleading and that chunking should be evaluated on full pipeline QA performance. Qu et al. (2024) question whether semantic chunking consistently

[1] Code and configs: https://github.com/kuromi1kow/chunkrag-course-project

justifies its computational cost, finding inconsistent gains across benchmarks. We build on these findings by comparing the four major chunking families in a single controlled pipeline and adding a post-hoc error analysis to separate retrieval failures from generation failures.

3 Methods

For each chunking strategy, the pipeline chunks each document, embeds every chunk with the same encoder, builds a FAISS dense index, retrieves the top 4 chunks for each question, and passes them to the same generator. The chunking function is the only variable.

3.1 Chunking Strategies

We implement four families:

- **Fixed token chunking:** `fixed_128`, `fixed_256`, `fixed_512`, with ∼15% overlap. Text is cut into fixed-length windows without regard for sentence or paragraph boundaries. Overlap helps preserve facts near boundaries, but coherent passages can still be split awkwardly.
- **Recursive chunking:** `recursive_256`, using LangChain's recursive splitter. This method preserves natural discourse units by applying an ordered list of separators: paragraph breaks first, then line breaks, then sentence-like punctuation, and only finally character-level cuts if needed. It keeps larger structures intact whenever they fit inside the token budget and backs off to finer splits only when necessary.
- **Sentence chunking:** `sentence_256`, using spaCy segmentation with greedy packing to a token budget. Documents are first segmented into sentences, then consecutive sentences are packed together until adding another would exceed the budget. This respects sentence boundaries but can still split related evidence across groups.
- **Semantic chunking:** `semantic_256`, splitting when inter-sentence cosine similarity drops below 0.72, with a minimum-length guard. Unlike fixed or structural methods, this approach uses embedding similarity to detect topical shifts, so chunk boundaries follow content structure rather than token counts.

3.2 Models and Configuration

The pipeline uses `all-MiniLM-L6-v2` for embedding, FAISS inner-product search over normalized embeddings for retrieval, and `Mistral-7B-Instruct-v0.3` served through vLLM on SCC as the generator. Prompting consists of a short extractive QA instruction plus a lightweight answer-compression step for verbose outputs. Generation is limited to 1024 input tokens and 48 new tokens. The broader codebase also supports BM25, hybrid retrieval, and reranking, but the analysis keeps the ablation focused on chunking. All hyperparameters are fixed in JSON configs and reused across runs.

3.3 Baseline

Our baseline is **parametric-only QA**: the generator answers without retrieved context. This tests whether retrieval is necessary and provides a lower bound.

3.4 Error Analysis

To attribute $\mathrm{EM} = 0$ failures, we assign each incorrect prediction to exactly one of seven categories in priority order (earlier categories take precedence when multiple could apply):

- *retrieval failure*: gold document absent from the top-4 chunks;
- *false refusal*: model outputs "unanswerable" despite adequate retrieval;
- *verbose extraction*: gold tokens are a strict subset of predicted tokens (right answer, extra words added);
- *terse extraction*: predicted tokens are a strict subset of gold tokens (correct but incomplete span);
- *close paraphrase*: $\mathrm{F1} \geq 0.6$, neither is a token-set subset of the other;
- *partial answer*: $0 < \mathrm{F1} < 0.6$; and
- *wrong answer*: $\mathrm{F1} = 0$.

For summary, categories 2–5 are grouped as *format or refusal*, all fixable by prompt or post-processing changes without touching the retriever or model weights.

System	EM	F1	Recall@4	Precision@4	Avg chunk tokens	# chunks
`parametric_only`	13.3	29.2	0.0	0.0	–	–
`fixed_128`	38.3	50.7	**95.0**	**77.1**	125.3	631
`fixed_256`	38.3	56.7	88.3	73.8	244.0	322
`fixed_512`	40.0	53.7	90.0	63.7	472.2	164
`recursive_256`	**41.7**	**60.6**	91.7	72.5	175.8	389
`sentence_256`	35.0	51.0	86.7	72.1	223.9	302
`semantic_256`	**41.7**	56.9	**95.0**	74.6	144.8	467

Table 1: Results on SQuAD v2. `recursive_256` is the strongest end-to-end setting.

System	EM	F1	Recall@4	Precision@4	Avg chunk tokens	# chunks
`parametric_only`	10.0	27.7	0.0	0.0	–	–
`fixed_128`	20.0	36.5	75.0	**42.5**	89.3	421
`fixed_256`	20.0	34.3	75.0	40.8	114.2	313
`fixed_512`	20.0	35.7	**76.7**	41.7	117.4	301
`recursive_256`	**26.7**	**42.6**	**76.7**	**42.5**	113.6	313
`sentence_256`	23.3	37.0	**76.7**	41.7	112.7	313
`semantic_256`	23.3	42.4	**76.7**	**42.5**	97.2	363

Table 2: Results on HotpotQA. Doc-level Recall@4 shows that recovering all supporting documents remains incomplete.

4 Experiments

4.1 Datasets

- **SQuAD v2:** 60 answerable validation questions sampled from a 500-example pool. Paragraphs sharing a title are concatenated into 35 article-level pseudo-documents to make chunking non-trivial.

- **HotpotQA distractor:** 30 validation questions with 300 documents built from distractor and supporting titles.

4.2 Evaluation

We report Exact Match (EM), token-level F1, Recall@4, and Precision@4 as quality metrics, and include average chunk size and chunk count as corpus statistics describing each chunker. Recall@4 is defined as the fraction of gold supporting documents that appear in the top-4 retrieved chunks. For SQuAD, each question has exactly one gold document, so Recall@4 is binary per question: 1 if any retrieved chunk came from that document, 0 otherwise. For HotpotQA, each question has two gold documents, so Recall@4 can take values 0, 0.5, or 1.0, making it a more informative measure of evidence coverage for multi-hop questions.

5 Results

All results below are from single runs on the SQuAD v2 subset ($n = 60$) and HotpotQA subset ($n = 30$).

The parametric-only baseline is weak on both datasets, confirming retrieval sensitivity. On SQuAD, `recursive_256` leads at 60.6 F1 and 41.7 EM, with `semantic_256` competitive at 56.9 F1 and tied for the best Recall@4 (95.0). `fixed_128` achieves the highest retrieval precision (77.1) but lower F1, suggesting that very small chunks recover evidence but may fragment it. Larger fixed windows (`fixed_512`) degrade F1 (53.7 vs. 56.7 for `fixed_256`), consistent with the hypothesis that broad chunks dilute relevant evidence with distractors; EM edges up slightly to 40.0, suggesting that length constraints hurt token-level overlap more than strict span matching.

On HotpotQA, `recursive_256` is best at 42.6 F1 and 26.7 EM, with `semantic_256` essentially tied on F1 at 42.4. Doc-level Recall@4 tops out at 76.7 rather than saturating, suggesting that evidence composition rather than chunk boundary placement is the main bottleneck once retrieval is only partially successful. A further sign that retrieval coverage and answer quality are decoupled: on SQuAD, `fixed_128` and `semantic_256` share the best Recall@4 (95.0) yet land 6.2 F1 points apart, and on HotpotQA four of six chunkers sit at Recall@4=76.7 with F1 spread from 34.3 to 42.6.

5.1 Error Analysis

Table 3 shows the coarse failure breakdown for `recursive_256`. The majority of EM = 0 cases are fixable format or refusal errors: 65.7% on SQuAD and 72.7% on HotpotQA. Retrieval accounts for only 14.3% of SQuAD failures and zero HotpotQA failures. The remaining 20–27% are genuine model errors where supporting evidence was present but the answer is substantially wrong. Figure 1 extends this view to all six chunkers and confirms the pattern: format-or-refusal errors dominate the EM = 0 bucket for every chunker on both datasets, retrieval failure is at most 20% on SQuAD and zero on HotpotQA, and genuine model errors stay in the 20–40% range.

Failure type	SQuAD		HotpotQA	
	n	%	n	%
Retrieval failure	5	14.3	0	0.0
Format or refusal (fixable)	23	65.7	16	72.7
Model error	7	20.0	6	27.3
Total EM = 0	35	100.0	22	100.0

Table 3: Coarse failure breakdown for `recursive_256`. "Format or refusal" groups verbose extraction, terse extraction, close paraphrases, and false refusals, all fixable without changing the retriever.

The fine-grained breakdown (Table 4) reveals that the dominant failure mode differs by dataset. On SQuAD, verbose extraction accounts for 40.0% of EM = 0 cases: the gold span is present inside the prediction but flanked by extra words. On HotpotQA, false refusals are the largest single category at 40.9%: Mistral outputs *unanswerable* for nearly four in ten of its mistakes. All four SQuAD false refusals had Recall@4 = 1.0, and four of nine HotpotQA false refusals likewise had full supporting-document coverage, confirming that the evidence was in the prompt each time the model refused. Annotated examples of each failure type are provided in Appendix A.

Failure type	SQuAD		HotpotQA	
	n	%	n	%
Retrieval failure	5	14.3	0	0.0
False "unanswerable"	4	11.4	9	40.9
Verbose extraction	14	40.0	5	22.7
Terse extraction	2	5.7	2	9.1
Close paraphrase	3	8.6	0	0.0
Partial answer	4	11.4	1	4.5
Wrong answer	3	8.6	5	22.7

Table 4: Fine-grained breakdown of EM = 0 predictions for `recursive_256`.

6 Discussion

Two findings stand out from this study. First, chunking strategy has a measurable, dataset-dependent effect on end-to-end QA quality. Second, the error analysis reveals that the headline EM scores understate true system quality: the majority of failures are prompt artifacts addressable without changing the retriever or model.

Chunking matters, and the best strategy depends on question type. On SQuAD, chunkers that preserve moderately large coherent spans (recursive, semantic) outperform both very small fixed windows and sentence-level segmentation. The F1 drop at `fixed_512` (despite a slight EM increase) confirms that overly broad chunks dilute token-level evidence with distractors, while `fixed_128`'s strong retrieval precision but weaker F1 suggests that very small chunks fragment evidence that should stay together. On HotpotQA, exact match is flatter across systems but F1 still varies, and the fact that fine-grained fixed chunks do not help multi-hop questions (despite their high individual-fact recall), suggesting that those questions require chunks large enough to preserve the local structure linking related facts. These patterns confirm that local factoid QA and multi-hop QA reward different chunking behaviors.

Most failures are fixable, and the dominant mode differs by dataset. On SQuAD, verbose extraction is the single largest failure category (40% of EM = 0): the model extracts the correct span but surrounds it with extra words, a systematic artifact of the current prompt's lack of explicit length constraints. On HotpotQA, false refusals dominate (41% of failures): Mistral outputs *unanswerable* even when all supporting evidence is present in the prompt. This is plausibly linked to multi-hop question complexity: synthesizing facts from two retrieved passages creates greater model uncertainty, and the current prompt's "say unanswerable if unsure" instruction provides an easy exit. Both failure modes are addressable through targeted prompt changes, without requiring better retrieval or a stronger model.

7 Conclusion

Chunking is a non-trivial experimental variable: across two QA benchmarks with different question structures, `recursive_256` is the most consistently strong chunker, and the parametric-

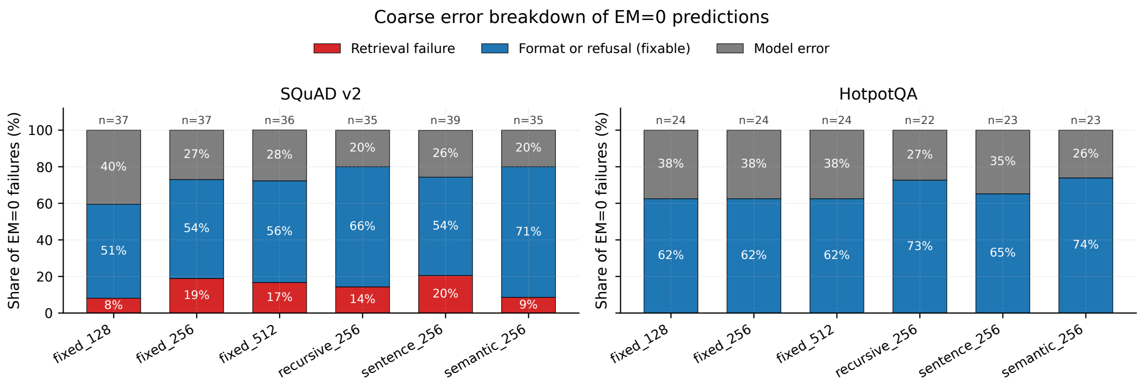

Figure 1: Coarse error breakdown of EM = 0 predictions by chunker. Each bar is normalised to 100%; the absolute number of EM = 0 predictions is annotated above each bar. "Format or refusal" aggregates verbose extraction, terse extraction, close paraphrases, and false refusals (categories fixable without changing the retriever or model). The fixable share never drops below 51% on any chunker–dataset cell.

only baseline falls well below every retrieval-based system. The dataset-level difference, where coherence-preserving chunkers dominate on single-hop SQuAD while the gap narrows on multi-hop HotpotQA, supports the hypothesis that the optimal chunking policy depends on question complexity.

The error analysis qualifies these conclusions in an important way. The majority of EM = 0 failures are fixable format or refusal errors rather than genuine retrieval or reasoning mistakes, and the dominant failure mode differs by dataset: verbose extraction on SQuAD and false refusals on HotpotQA. These findings suggest that dataset-specific prompt improvements are the highest-leverage next step, and that once prompt issues are resolved, the chunking comparison can be extended cleanly to BM25, hybrid, and reranked retrieval settings already implemented in the codebase.

Author Contributions

Assylkhan Geniyat led the SCC `vLLM` deployment (`Mistral-7B` + endpoint tunneling) and the SCC rigorous experiment sweep, and contributed to the report writing. **Pablo Bello** built the chunking, retrieval, and evaluation pipeline (`src/chunkrag`), the error-analysis script and figures, and the ACL report draft. **Jaime Bernal** implemented the Streamlit demo dashboard and the local OpenWebUI integration, and ran the parametric-only and quickstart baselines. **Batyrkhan Baimukhanov** contributed the Chonkie auxiliary comparison, dataset preprocessing utilities, and the qualitative failure-example annotations in Appendix A. All authors participated in discussions of methodology and reviewed the final manuscript.

8 Limitations

Strict metrics and answer-format mismatch. EM and F1 penalize surface-form mismatches even when semantic content is correct. The error analysis quantifies the impact: 40% of SQuAD EM = 0 cases are verbose extractions where the gold span is present inside the prediction, making metric mismatch the single largest source of the SQuAD score gap.

Uniform prompting across datasets. The system uses a single extractive QA prompt for both datasets. HotpotQA's multi-hop reasoning would benefit from a dataset-specific prompt that softens the unanswerable instruction and includes document titles so the model can track which passage each fact comes from.

Limited post-processing. The existing answer normalizer handles common formatting issues but does not cover all observed mismatch patterns such as determiners, parenthetical clarifications, and unit formatting.

Small evaluation subsets. The subsets (60 and 30 questions) are sufficient to reveal broad trends, but the ranking among chunkers could shift with larger samples.

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A Failure Examples

Four representative annotated failures from `recursive_256`, one per category in the coarse breakdown of Table 3: *retrieval failure*, *false refusal* (the dominant fixable mode on HotpotQA), *verbose extraction* (the dominant fixable mode on SQuAD), and *wrong answer* (genuine model error). Predictions are verbatim.

Retrieval failure (SQuAD) *Q:* What entity owns V/Line?
Gold: Victorian Government *Pred: unanswerable*
F1: 0.00 *Recall@4:* 0.00

The retriever returned documents about unrelated topics; the gold article was never in the prompt. This failure is unrecoverable by prompt changes alone; it requires better retrieval.

False refusal (HotpotQA) *Q:* Robert Earl Holding owned an oil company that was originally founded by who?
Gold: Harry F. Sinclair *Pred: unanswerable*
F1: 0.00 *Recall@4:* 1.00

All four supporting documents were present in the prompt. The model declined to answer despite having the evidence, a pure prompt-tone failure fixable by softening the unanswerable instruction.

Verbose extraction (SQuAD) *Q:* What is thought to have happened to the *Y. pestis* that caused the Black Death?
Gold: may no longer exist *Pred:* Y. pestis may no longer exist
F1: 0.80 *Recall@4:* 1.00

The gold span is present verbatim inside the prediction, preceded by the question subject. EM fails entirely; F1 is penalised by the extra tokens. Tightening the prompt to suppress subject echoing would recover this case.

Wrong answer (HotpotQA) *Q:* Which mountain is taller, Gasherbrum II or Langtang Ri?
Gold: Gasherbrum II *Pred:* taller than Langtang Ri
F1: 0.00 *Recall@4:* 1.00

Both supporting articles were retrieved. The model named the wrong mountain with zero token overlap against the gold. This represents a genuine reasoning failure that cannot be attributed to format or retrieval.